📷 Angstrong Nuwa-HP60C 3D Depth Camera Introduction and Usage

1. Course Overview

This course will introduce the basic characteristics of the **Angstrong Nuwa-HP60C 3D Structured Light Depth Camera** equipped on the A1 Pro robot, and demonstrate how to use ROS 2 driver to start the camera node, use `rqt_image_view` and `RViz2` to view RGB color stream, Depth stream, IR infrared images, and 3D spatial dense point cloud in real-time.

🎯 Course Objectives

1. Understand the structured light measurement principle and core hardware parameters of the Angstrong Nuwa-HP60C depth camera
2. Master how to use `ascamera` to start the depth camera driver node
3. Master how to subscribe and visualize RGB color image, 16-bit depth map, and 3D point cloud in ROS 2 environment

> **Important**
>
> This course is only applicable to users with the **Angstrong Nuwa-HP60C Depth Camera** version.

---

2. Camera Hardware Introduction



The Angstrong Nuwa-HP60C is an industrial-grade micro 3D structured light depth camera, consisting of VCSEL laser speckle projector, infrared receiver, and high-resolution RGB sensor:

- **Measurement Range**: 0.2m ~ 4m
- **RGB**: 1280×720 / 640×480 @ 30FPS
- **Depth**: 640×480 @ 30FPS, 16-bit millimeter-level accuracy
- **Field of View**: H73.8° × V51.7°
- **Interface**: Type-C USB 2.0 / 3.0

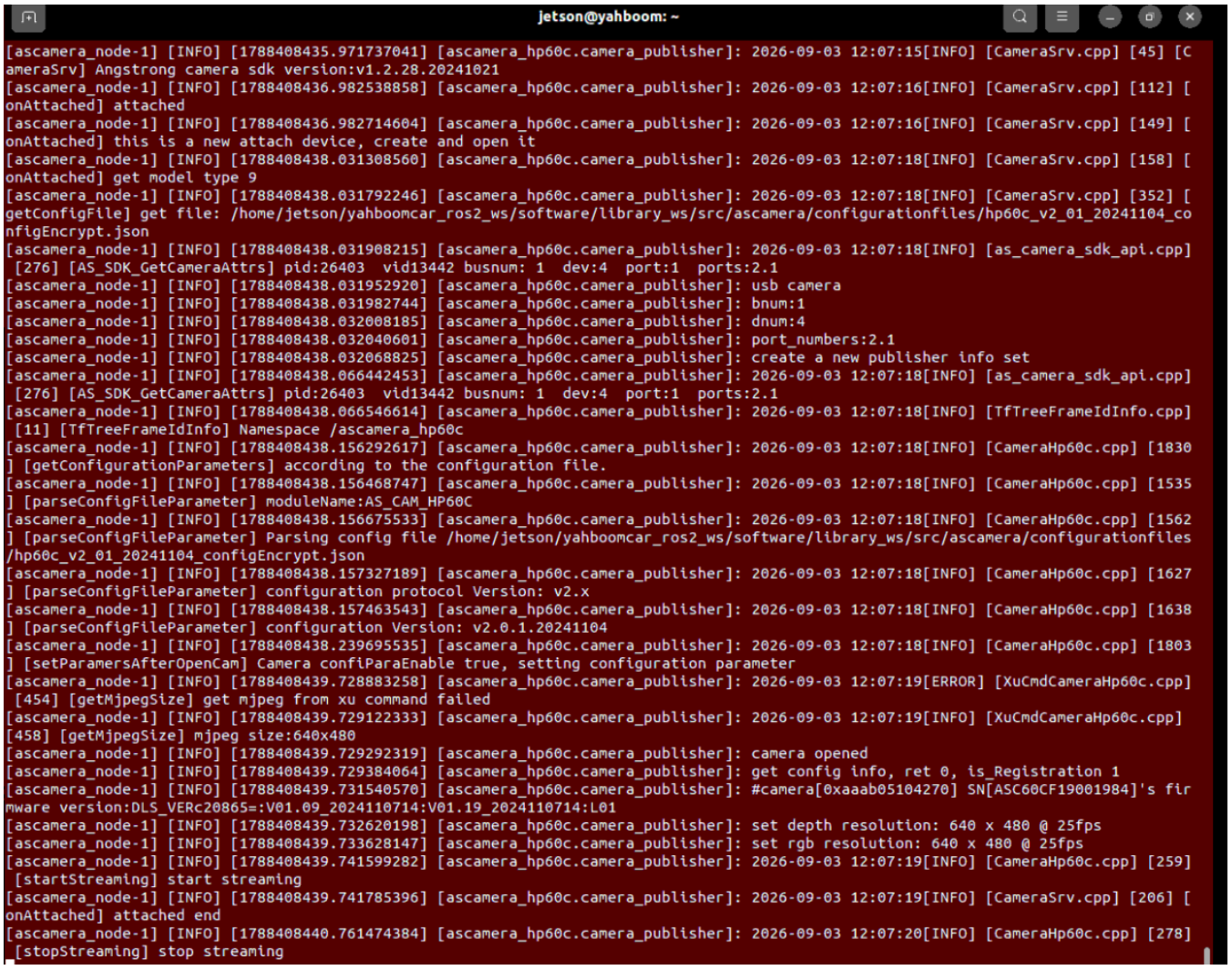
---

3. Preparation and Driver Startup

3.1 Start Camera Driver Individually

Open the terminal and run the following command to start the camera low-level driver:

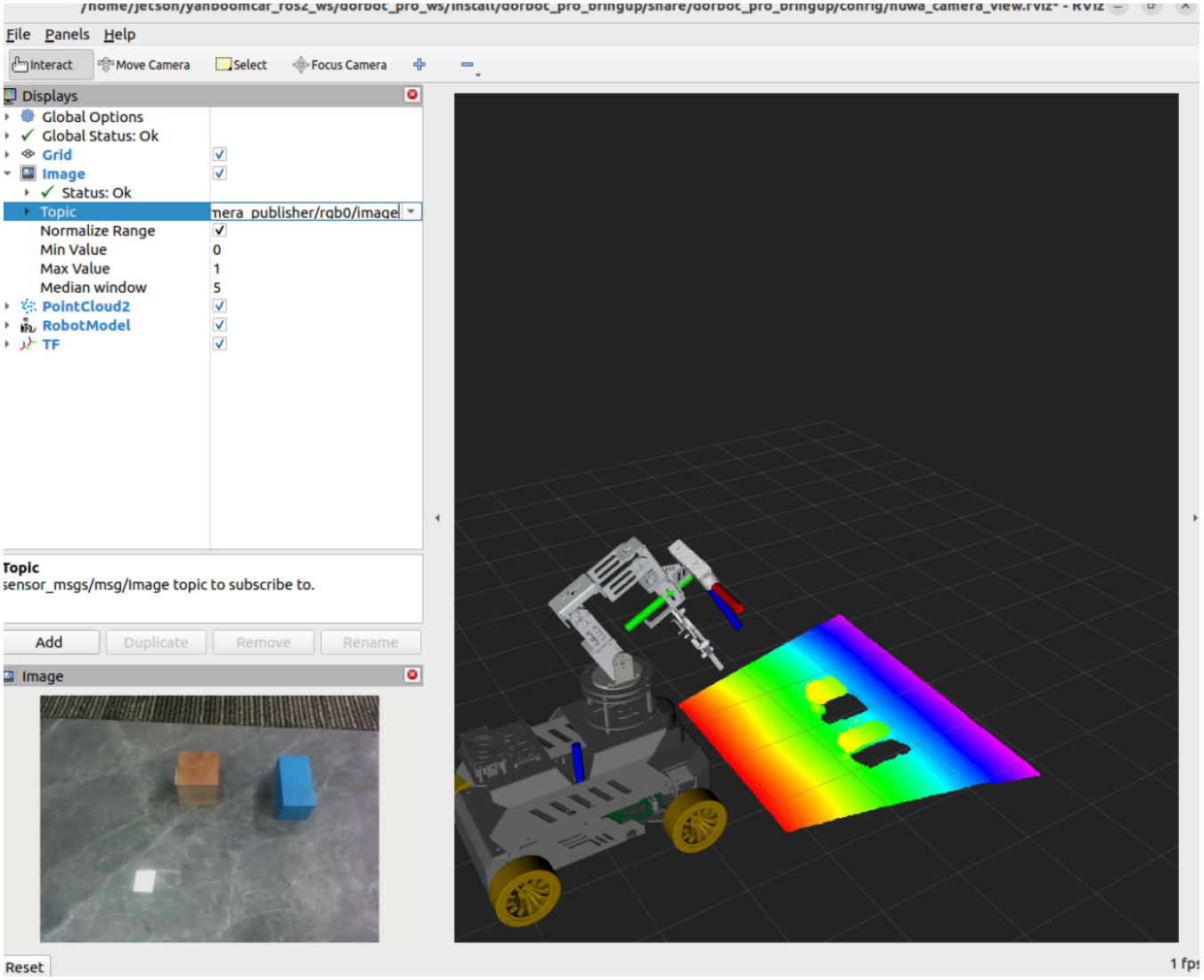
```
ros2 launch ascamera hp60c.launch.py
```



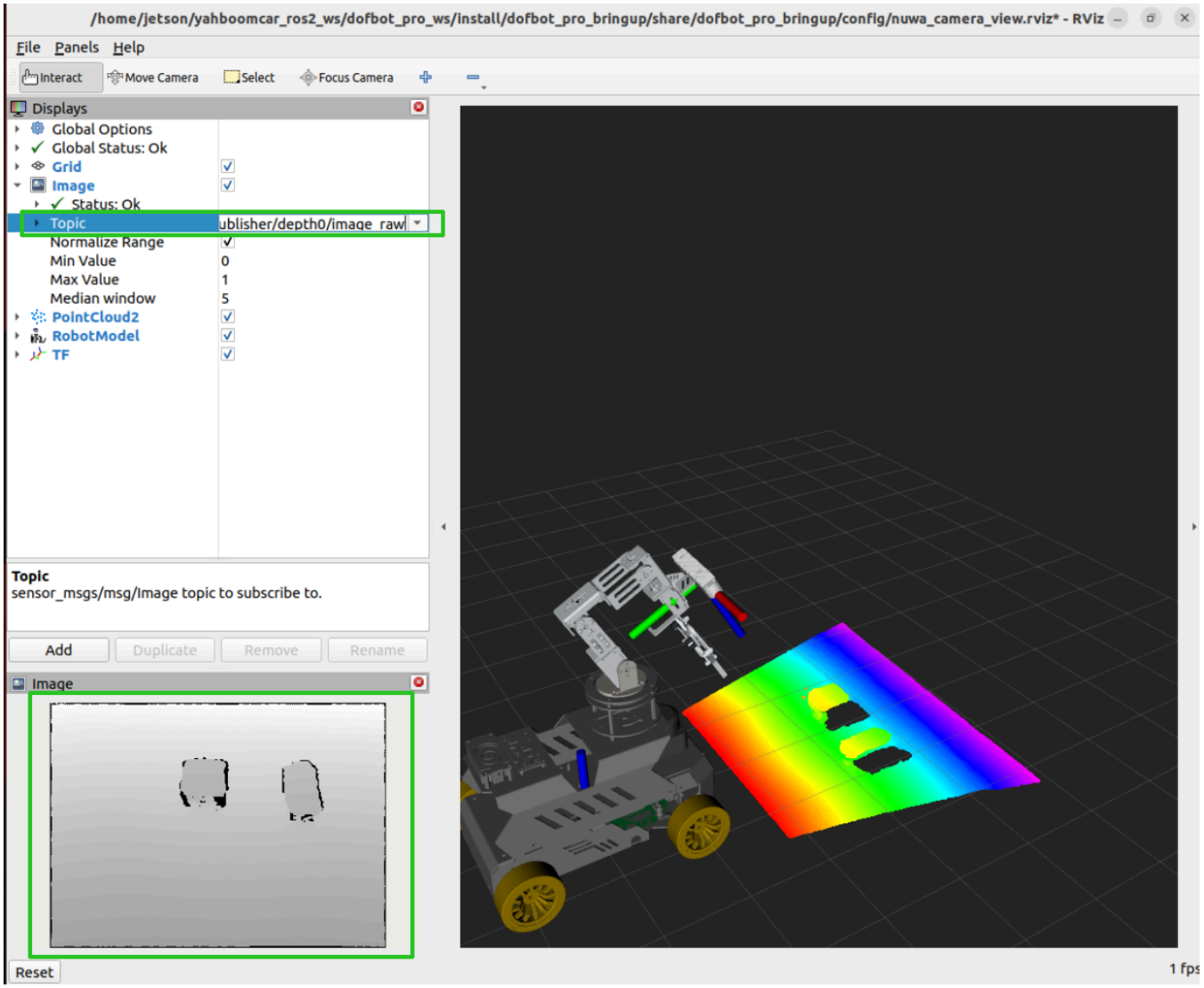
3.2 Start Camera Driver and RViz Visualization Integrated Environment

If you need to start the driver and automatically pop up `RViz2` point cloud monitoring and topic monitoring:

```
ros2 launch dofbot_pro_bringup nuwa_camera_view.launch.py
```



- Subscribe to depth image, select `/ascamera_hp60c/camera_publisher/depth0/image_raw` depth image topic



4. Hands-on Demo and Topic Monitoring

4.1 View Camera Node and Topic Information

Execute in the terminal:

```
ros2 node info /ascamera_hp60c/camera_publisher
```

The published topics are shown in the figure below:

```
/ascamera_hp60c/camera_publisher
  Subscribers:
    /parameter_events: rcl_interfaces/msg/ParameterEvent
  Publishers:
    /ascamera_hp60c/camera_publisher/depth0/camera_info: sensor_msgs/msg/CameraInfo
    /ascamera_hp60c/camera_publisher/depth0/image_raw: sensor_msgs/msg/Image
    /ascamera_hp60c/camera_publisher/depth0/points: sensor_msgs/msg/PointCloud2
    /ascamera_hp60c/camera_publisher/rgb0/camera_info: sensor_msgs/msg/CameraInfo
    /ascamera_hp60c/camera_publisher/rgb0/image: sensor_msgs/msg/Image
    /parameter_events: rcl_interfaces/msg/ParameterEvent
    /rosout: rcl_interfaces/msg/Log
    /tf: tf2_msgs/msg/TFMessage
```

The provided services are shown in the figure below:

```
  Service Servers:
    /ascamera_hp60c/camera_publisher/describe_parameters:
rcl_interfaces/srv/DescribeParameters
    /ascamera_hp60c/camera_publisher/get_parameter_types:
rcl_interfaces/srv/GetParameterTypes
    /ascamera_hp60c/camera_publisher/get_parameters: rcl_interfaces/srv/GetParameters
    /ascamera_hp60c/camera_publisher/list_parameters: rcl_interfaces/srv/ListParameters
    /ascamera_hp60c/camera_publisher/set_parameters: rcl_interfaces/srv/SetParameters
    /ascamera_hp60c/camera_publisher/set_parameters_atomically:
rcl_interfaces/srv/SetParametersAtomically
```